

Ali Habiboglu

University of Arizona

Email: alitahahabiboglu@arizona.eduWebsite: alitahahabiboglu.github.io/ali/**Research Interests**

• First Wall and Plasma-facing Materials for Fusion Energy • High-Entropy Thin Films • Graphene Nanoribbons (GNRs) • Condensed Matter Physics • Metal/Oxide Thin-Film Growth and Epitaxy • Physical Vapor Deposition (PVD) and Magnetron Sputtering • Thin-film Characterization under Extreme Environments • Device Physics • Semiconductors • Nanoelectronics • Microelectronics • 2D Materials • Quantum Materials & Devices • Magnetic Materials • Memory/Logic Technologies • Transistors • Spintronics • Quantum Sensing • Magnetic Sensors • Magnetic Tunnel Junctions (MTJs) • Low-Energy High Performance Computing • In-Memory Computing

Academic Appointments and Education• **University of Arizona, Tucson**

Tucson, AZ

Provost's Postdoctoral Fellow Department of Materials Science and Engineering

(2026 - Present)

Mentor: Professor Zafer Mutlu

Doctor of Philosophy in Physics with Materials Science and Engineering Minor

2017 - 2026

Thesis: Advanced Magnetic Tunnel Junctions with Novel Electrodes and Barriers

Advisor: Professor Weigang Wang

Master of Science in Physics

2017 - 2022

Advisor: Professor Weigang Wang

• **Bogazici University**

Istanbul, Türkiye

Bachelor of Science in Physics

2011 - 2016

Advisor: Professor Ozhan Ozatay

Research Experience• **University of Arizona**

Tucson, Arizona

Graduate Research Assistant, Spin Applications Lab, Physics Department

2017 - 2026

Supervisor: Professor Weigang Wang• **Sabancı University**

Istanbul, Türkiye

Visiting Undergraduate Researcher, Sabancı University Nanotechnology Research & Application Center 2014 - 2016**Supervisor:** Doctor Cenk Yanik• **Bogazici University**

Istanbul, Türkiye

Undergraduate Research Assistant, Magnetic Materials and Memory Lab, Physics Department

2013 - 2017

Supervisor: Professor Ozhan Ozatay**Publications**

For an up-to-date publication list, please see my Google Scholar profile. https://scholar.google.com/citations?user=w_5i1ywAAAAJ&hl=en

Journal Papers

1. A. Habiboglu, B. Larsen, C. Webb, K. Svoboda, Y. Choudhary, M. Elektiar, C. Sullivan, J. O'Brien, C. Eckel, S. Hajirnis, S. Majetich, T. Adegbiya, S. Zhang, E. Tsymbal, W. Wang, "Momentum Dependent Spin Polarization based Room-temperature Tunneling Magnetoresistance in PtMn_x based hybrid Antiferromagnetic Tunnel Junctions," (Under preparation).
2. A. Habiboglu, B. Larsen, J. Zhang, M. Chen, P. Khanal, B. Zhou, J. O'Brien, C. Eckel, S. Hajirnis, W. Wang, "Implementation of Antiferromagnetic Barrier with High Tunneling Magnetoresistance: A Pathway Toward Magnon-Assisted Spin-Transfer Torque Switching," IEEE Transactions on Magnetics, (Under review).
3. Y. Zhang, M. Xu, B. Zhou, C. Eckel, S. Ghosh, H. Yun, A. Habiboglu, D. Lyu, D. B. Gopman, J.-P. Wang, K. A. Mkhoyan, W. Wang, "Breaking voltage-controlled magnetic anisotropy-magnetoresistance trade-off for ultralow-energy switching in magnetic tunnel junctions," Science Advances, vol. 12, eaef6328, 2026.
4. P. Khanal, B. Zhou, H. Almasi, A. Habiboglu, M. Andrade, J. O'Brien, A. Enriquez, C. Eckel, C. Mastrangelo, W.-G. Wang, "Interplay of symmetry-conserved tunneling, interfacial oxidation and perpendicular magnetic anisotropy in CoFeB/MgO-based junctions," Journal of Physics D: Applied Physics, vol. 59, no. 24, p. 245001, 2026.
5. Y. Lv, B. R. Zink, R. P. Bloom, H. Cilasun, P. Khanal, S. Resch, Z. Chowdhury, A. Habiboglu, W. Wang, S. S. Sapatnekar, U. Karpuzcu, J.-P. Wang, "Experimental demonstration of magnetic tunnel junction-based computational random-access memory," npj Unconventional Computing, vol. 1, no. 1, 2024.

6. D. Lyu, J. E. Shoup, A. T. Habiboglu, Q. Jia, P. Khanal, B. R. Zink, Y. Lv, B. Zhou, D. B. Gopman, W. Wang, J.-P. Wang, "L1₀ FePd-based perpendicular magnetic tunnel junctions with 65% tunnel magnetoresistance and ultralow switching current density," *AIP Advances*, vol. 14, 025019, 2024.
7. A. Habiboglu, Y. Chandak, P. Khanal, B. Larsen, B. Zhou, C. Eckel, J. Cutshall, K. Warrilow, J. O'Brien, B. Hong, J. R. Schaibley, B. J. Leroy, W. Wang, "The impact of boron doping in the tunneling magnetoresistance of Heusler alloy Co₂FeAl," *Journal of Applied Physics*, vol. 134, no. 24, 2023.
8. M. Dc, D.-F. Shao, V. D.-H. Hou, A. Vailionis, P. Quarterman, A. Habiboglu, M. B. Venuti, F. Xue, Y.-L. Huang, C.-M. Lee, M. Miura, B. Kirby, C. Bi, X. Li, Y. Deng, S.-J. Lin, W. Tsai, S. Eley, W.-G. Wang, J. A. Borchers, E. Y. Tsymbal, S. X. Wang, "Observation of anti-damping spin-orbit torques generated by in-plane and out-of-plane spin polarizations in MnPd₃," *Nature Materials*, vol. 22, no. 5, pp. 591–598, 2023.
9. B. Zhou, P. Khanal, O. J. Benally, D. Lyu, D. B. Gopman, A. Enriquez, A. Habiboglu, K. Warrilow, J.-P. Wang, W.-G. Wang, "Perpendicular magnetic anisotropy, tunneling magnetoresistance and spin-transfer torque effect in magnetic tunnel junctions with Nb layers," *Scientific Reports*, vol. 13, 3454, 2023.
10. P. Khanal, B. Zhou, M. Andrade, C. Mastrangelo, A. Habiboglu, A. Enriquez, D. Fox, K. Warrilow, W.-G. Wang, "Enhanced magnetoresistance in perpendicular magnetic tunneling junctions with MgAl₂O₄ barrier," *Journal of Magnetism and Magnetic Materials*, vol. 563, 169914, 2022.
11. P. Khanal, B. Zhou, M. Andrade, Y. Dang, A. Davydov, A. Habiboglu, J. Saidian, A. Laurie, J.-P. Wang, D. B. Gopman, W. Wang, "Perpendicular magnetic tunnel junctions with multi-interface free layer," *Applied Physics Letters*, vol. 119, no. 24, 2021.
12. M. Xu, M. Li, P. Khanal, A. Habiboglu, B. Insana, Y. Xiong, T. Peterson, J. C. Myers, D. Ortega, H. Qu, C. L. Chien, W. Zhang, J.-P. Wang, W. Wang, "Voltage-controlled antiferromagnetism in magnetic tunnel junctions," *Physical Review Letters*, vol. 124, no. 18, 187701, 2020.
13. V. Karakas, A. Gokce, A. T. Habiboglu, S. Arpacı, K. Ozbozduman, I. Cinar, C. Yanik, R. Tomasello, S. Tacchi, G. Siracusano, M. Carpentieri, G. Finocchio, T. Hauet, O. Ozatay, "Observation of magnetic radial vortex nucleation in a multilayer stack with tunable anisotropy," *Scientific Reports*, vol. 8, 7180, 2018.

Patent Application

14. A. Habiboglu, Y. Cheng, S. Zhang, W. Wang, "Magnetic tunneling junctions with a magnetic barrier," US Patent US-12096698-B2, 2024.

Professional Presentations

Invited Talks/Posters

1. A. Habiboglu, B. Larsen, J. Zhang, M. Chen, P. Khanal, B. Zhou, J. O'Brien, C. Eckel, S. Hajirnis, W. Wang, "Implementation of Antiferromagnetic Barrier with High Tunneling Magnetoresistance: A Pathway Toward Magnon-Assisted Spin-Transfer Torque Switching," The Magnetic Recording Conference, Poster Presentation, 2026.
2. A. Habiboglu, K. Yumigeta, M. Yusufoglu, Z. Mutlu, "Design of Self-Healing First-Wall Coatings Using High-Entropy Alloys with Embedded Graphene Nanoribbon Radiation Diagnostics," Arizona Livermore Days Workshop, Oral Presentation, 2026.
3. A. Habiboglu, B. Larsen, C. Webb, K. Svoboda, Y. Choudhary, M. Elektiar, C. Sullivan, S. Majetich, T. Adegbiya, S. Zhang, E. Tsymbal, W. Wang, "Ultra-fast energy-efficient antiferromagnetic tunnel junctions for MRAM applications," Future of Semiconductors (FuSe2), NSF Headquarters, Poster Presentation, 2025.
4. A. Habiboglu, "Wide Range Magnetic Sensors," SEMICON West, Participation sponsored by the National Science Foundation (NSF) I-Corps Program, 2025.
5. A. Habiboglu, B. Larsen, R. Dierdorff, K. Musarandega, B. Hong, B. Zhou, J. O'Brien, C. Eckel, W. Wang, "Fabrication and Characterization of CMOS-Compatible Antiferromagnetic Tunnel Junctions," APS March Meeting, Oral Presentation, 2025.
6. A. Habiboglu, B. Hong, P. Khanal, B. Larsen, B. Zhou, J. O'Brien, C. Eckel, K. Warrilow, A. Enriquez, W. Wang, "Magnetic Tunnel Junctions with non-collinear Antiferromagnets," FUSENANO, Poster Presentation, 2024.
7. A. Habiboglu, B. Hong, P. Khanal, B. Larsen, B. Zhou, J. O'Brien, C. Eckel, K. Warrilow, A. Enriquez, W. Wang, "Bias Dependence of Tunneling Magnetoresistance in Magnetic Tunnel Junctions with Antiferromagnetic Pt_xMn_{1-x}," MMM Conference, Oral Presentation, 2023.
8. A. Habiboglu, Y. Chandak, P. Khanal, B. Zhou, C. Eckel, J. O'Brien, W. Wang, "Impact of Boron doping to the tunneling magnetoresistance of Heusler alloy Co₂FeAl," APS March Meeting, Oral Presentation, 2023.

9. A. T. Habiboglu, V. Karakas, M. Mete, A. Coskuner, Y. B. Kalyoncu, A. Gokce, A. Giordano, M. Carpentieri, F. Celegato, P. Tiberto, G. Finocchio, O. Ozatay, "Static and Dynamic Properties of Magnetic Antivortices in Asteroid-Shaped Permalloy Nanomagnets," APS March Meeting, Oral Presentation, 2015.

Contributed Presentations

10. R. M. Dierdorff, A. Habiboglu, B. Larsen, K. Musarandega, B. Zhou, J. O'Brien, B. Hong, W. Wang, "Comparing Spin-Orbit Torque Switching in All-Ferromagnetic and All-Antiferromagnetic Tunnel Junctions," American Physical Society (APS) March Meeting, Oral Presentation, 2025.
11. D. Gopman, J. Shoup, D. Lyu, A. Habiboglu, D. Huang, Q. Jia, P. Khanal, B. Zink, Y. Lv, B. Zhou, X. Wang, W. Wang, J.-P. Wang, "Advanced Magnetic Tunnel Junctions for In- and Near-Memory Compute," IEEE-NANO, Oral Presentation, 2025.
12. D. Lyu, D. Zhang, B. R. Zink, J. E. Shoup, M. Bapna, A. Habiboglu, W. Jiang, Y. Lv, D. Sousa, Y.-C. Liao, Q. Jia, P. Khanal, Z. Zhao, O. J. Benally, H. Li, P. Sahu, B. Zhou, A. Naeemi, D. B. Gopman, W. Wang, T. Low, S. Majetich, J.-P. Wang, "Investigation of Materials, Stacks, and Energy-Efficient Switching of Voltage-Controlled Exchange Coupling Perpendicular Magnetic Tunnel Junctions," American Physical Society (APS) March Meeting, Oral Presentation, 2024.
13. P. Khanal, B. Zhou, M. Andrade, C. Mastrangelo, A. Habiboglu, A. Enriquez, D. Fox, K. Warrilow, W.-G. Wang, "Tuning the oxygen ratio in perpendicular magnetic tunnel junctions with MgAl_2O_4 barriers," American Physical Society (APS) March Meeting, Oral Presentation, 2023.
14. B. Zhou, P. Khanal, O. J. Benally, D. Lyu, D. B. Gopman, A. Enriquez, A. Habiboglu, K. Warrilow, J.-P. Wang, W.-G. Wang, "Perpendicular magnetic anisotropy, tunneling magnetoresistance and spin-transfer torque effect in magnetic tunnel junctions with Nb layers," American Physical Society (APS) March Meeting, Oral Presentation, 2023.
15. C. Eckel, P. Khanal, A. Habiboglu, B. Zhou, F. MahdikhanySarvejahany, J. Schaibley, W. Wang, "Interlayer exchange coupling in the free layers of perpendicular magnetic tunnel junctions," American Physical Society (APS) March Meeting, Oral Presentation, 2022.
16. M. Andrade, P. Khanal, A. Habiboglu, B. Zhou, J. O'Brien, W. Wang, "Spin Dependent Transport in Magnetic Tunneling Junctions," American Physical Society (APS) March Meeting, Oral Presentation, 2020.
17. W. Wang, A. Habiboglu, P. Khanal, B. Zhou, M. Xu, H. Almasi, "Perpendicular magnetic tunneling junctions with above 500% TMR: MgO and beyond – contributed to the optimization of MTJ structure and presented BN and SrTiO_3 barriers," Semiconductor Research Corporation (SRC), Annual Review Meeting, 2019.
18. K. Ozbozduman, V. Karakas, S. Arpacı, A. T. Habiboglu, A. Gokce, A. Giordano, F. Celegato, P. Tiberto, G. Finocchio, G. Aktas, O. Ozatay, "Spin Torque induced anti-vortex excitations," American Physical Society (APS) March Meeting, Oral Presentation, 2016.

Awards, Honors, & Recognitions

1. IEEE Neil Smith Award for Contributions to the Understanding of Magnetic Phenomena in Materials and Devices, "Implementation of Antiferromagnetic Barrier with High Tunneling Magnetoresistance: A Pathway Toward Magnon-Assisted Spin-Transfer Torque Switching," The Magnetic Recording Conference (TMRC), 2026
2. Provost's Postdoctoral Fellowship, University of Arizona (2026)
3. Fanfare Graduate Travel Award, Fan Family; Department of Physics, University of Arizona (2025)
4. NSF Innovation-Corps (I-Corps) Program: I-Corps Immerse travel support to attend SEMICON WEST (2025)
5. Recognition of patent at the 2024 CoS Teaching, Advising, and Entrepreneurship Award Reception, College of Science, University of Arizona (2024)
6. NSF Innovation-Corps (I-Corps) Program: Presented technology concept "Wide-Range Magnetic Sensor" and received a \$5,000 grant for customer discovery and market-fit exploration (2023)
7. Third Place, Türkiye Rowing National Championships – Coxswain, Bogazici University team (2012)

Teaching

Instructor & Curriculum Developer (Semiconductor Fabrication and Design) (2024 - 2025)

1. Co-developed and taught **two graduate-level courses** covering semiconductor fabrication, device physics, and advanced logic circuit design.
2. Designed one curricula for **out-of-field faculty** and one for **in-field faculty** to be narrated by an AI supported avatar.

3. Delivered live Q&A and lab sessions to approximately **40 professors and PhD students** from **Tecnológico de Monterrey** (ranked #2 in Mexico).
4. Collaborated with engineers across multiple disciplines on course design and delivery.

Lab Instructor & Graduate Teaching Assistant

(2017 - 2018 & 2021 - 2022)

Department of Physics, University of Arizona

5. **Electricity and Magnetism Laboratory (Physics 241):** Instructor for two sections (~20 students each).
6. **Mechanics Laboratory (Physics 141):** Instructor for one section (~20 students).
7. **Introductory Physics (Physics 102):** Led discussion sections for three classes (>30 students each).
8. Graded exams and homework for **three undergraduate physics courses**.

Undergraduate Physics & Mathematics Tutor

(2011 - 2017)

9. Completed **400+ hours** of tutoring in Physics and Mathematics (101, 102, 201, 202 levels).

Research Mentoring

Ph.D. Student, University of Arizona

1. Brecken Larsen, Physics Department, University of Arizona 2022–2024
 - Project: Antiferromagnetic Tunnel Junctions (AFMTJs) for low-energy and high-frequency switching (NSF FuSe2).
 - Tasks: Thin-film deposition (PVD), MTJ device fabrication, and process optimization.
 - Outcomes: Fabrication and testing of AFMTJs; contributed to the group becoming the 4th worldwide to demonstrate AFMTJ operation.

Undergraduate Students

2. Mile Chen, Physics Department, University of Arizona 2025–2026
 - Project: Magnon-assisted spin-transfer torque switching in NiO-based MTJs (NSF).
 - Tasks: Magnetic field dependent measurements and data visualization.
 - Outcomes: Contributed to conference poster and 2-page abstract presented in TMRC.
3. Thomas D. Bradstreet, Physics Department, University of Arizona 2025–2026
 - Project: Mn₃Ga-based Antiferromagnetic Tunnel Junctions (AFMTJs) (NSF FuSe2).
 - Tasks: Electrical testing and data analysis of Spin-Orbit Torque (SOT) switching of AFMTJs.
 - Outcomes: Created an article for his final project in Physics program.
4. Ruth M. Dierdorff, Physics Department, University of Arizona 2024–2025
 - Project: Mn₃Pt-based Antiferromagnetic Tunnel Junctions (AFMTJs) (NSF FuSe2).
 - Tasks: Electrical testing and data analysis of Spin-Orbit Torque (SOT) switching of AFMTJs.
 - Outcomes: Presented research through the UROC Program and at the APS March Meeting 2025.
5. Roman Valle, Physics Department, University of Arizona 2024–2025
 - Project: Investigation of the magnetic behavior of TaCrP.
 - Tasks: Magnetic characterization and analysis.
 - Outcomes: Completed an ASEMS STARS undergraduate research project.
6. Kudzai Musarandega, Physics Department, University of Arizona 2023–2024
 - Project: Exploring the bias-dependence of PtMn_x-based tunnel junctions (NSF).
 - Tasks: Measuring voltage-dependent transport on tunnel junctions grown under varied conditions.
 - Outcomes: Voltage-driven enhancement of tunneling anisotropic magnetoresistance (TAMR) is analyzed.
7. Kennedy Warrilow, Physics Department, University of Arizona 2022–2023
 - Project: Magnon-assisted spin-transfer torque switching in NiO-based MTJs (NSF).
 - Tasks: Electrical testing, data analysis, and optimization of MTJ performance.

- Outcomes: Achieved record (26%) tunneling magnetoresistance for NiO-based MTJs.
8. Yash Chandak, Physics Department, University of Arizona 2021–2022
 - Project: Boron-doped Heusler alloy (Co_2FeAl) magnetic tunnel junctions (SRC).
 - Tasks: Device testing, transport measurements, and data analysis.
 - Outcomes: Contributed to a 50% increase in TMR and a peer-reviewed journal publication.
 9. Carter Eckel, Physics Department, University of Arizona 2021–2022
 - Project: Exchange-coupled magnetic tunnel junctions (Purdue University collaboration).
 - Tasks: Multilayer thin-film fabrication using PVD and photolithography.
 - Outcomes: Demonstrated controlled interlayer exchange coupling in MTJ devices.
 10. Christopher Mastrangelo, Physics Department, University of Arizona 2020–2021
 - Project: Voltage-controlled magnetic anisotropy using SrTiO_3 barriers (SRC).
 - Tasks: Development of a high-speed electrical measurement system.
 - Outcomes: Increased measurement throughput by a factor of six.
 11. Kaan Ozbozduman, Physics Department, Boğaziçi University 2015–2017
 - Project: Observation of magnetic skyrmions (radial vortices).
 - Tasks: Setting up electrical measurement systems and programming their software using LabVIEW.
 - Outcomes: Co-authored a publication confirming theoretically predicted radial vortex states.

High School Students

12. Jerrick Zhang 2024–2026
 - Project: Magnon-assisted spin-transfer torque switching in NiO-based MTJs (NSF).
 - Tasks: Low-T/RT/High-T DC/RF TMR measurements and data visualization.
 - Outcomes: Contributed to an article and TMRC conference presentation.
13. Brady Hong 2021–2023
 - Project: Boron-doped Heusler alloy MTJs.
 - Tasks: Assisted with electrical measurements and data processing.
 - Outcomes: Contributed to peer-reviewed publications and conference presentations.
14. Makeen Shafer 2021–2022
 - Project: Magnetic characterization of synthesized crystalline materials.
 - Tasks: Magnetometry measurements and data interpretation.
 - Outcomes: Hands-on training in experimental magnetic characterization.
15. Tarik Baysal 2021–2022
 - Project: Boron-doped Heusler alloy MTJs.
 - Tasks: Assisted with electrical testing and data analysis.
 - Outcomes: Supported experimental validation of enhanced TMR.

Professional Activities

Journal Review

1. Applied Physics Letter (AIP Publishing, January 2025 & June 2026)
2. Transactions on Magnetics (IEEE, December 2024 & August 2026)
3. Applied Electronic Materials (ACS, August 2024)

New Course Development

1. Developed **two graduate-level courses** in Semiconductor Fabrication and Design, tailored for **out-of-field** and **in-field faculty**.

2. Designed **2 curricula**, including lectures, problem sets, lab exercises, and assessment materials.
3. Created **online course videos** to support remote learning and interactive content delivery.
4. Developed and delivered **live laboratory sessions** to reinforce conceptual understanding.
5. Collaborated with engineers from multiple disciplines to deliver a **complete course** for graduate students and faculty starting from device physics to logic circuit design.
6. Trained **over 40 participants** (professors and PhD students) over two semesters, with online courses and live laboratory/problem-solving sessions.

Laboratory Infrastructure, System Integration, and Upgrades

1. **Designed and built** an ultra-high-vacuum rapid thermal annealing (**UHV-RTA**) system, enabling controlled post-growth thermal processing for thin-film and devices.
2. **Refurbished multiple PVD systems**, with 5-gun, 4-gun, and custom 2-gun configurations, restoring full operational capability and expanding experimental flexibility.
3. Developed a fast DC electrical measurement platform using Keithley 2400 and custom LabVIEW VIs, achieving 6× improvement in measurement speed.
4. **Designed and fabricated** custom deposition holders to improve the fabrication throughput in **PVD** systems.
5. Performed **maintenance, diagnostics, and troubleshooting** across major characterization platforms, including XRD, MOKE, PPMS, wire bonding, and related systems.
6. **Refurbished a mask aligner** and conducted a process optimization for photolithography, etching, and thin-film deposition workflows.
7. **Engineered a cooling solution** for PPMS compressors to ensure stable operation under desert environmental conditions.

Research Instrumentation Training and Operational Documentation

1. Prepared comprehensive manuals for fabrication and characterization systems, including: PVD Magnetron Sputtering, Rapid Thermal Annealing (RTA), and Mask Aligner for both positive and negative photolithography.
2. Developed training guides for thin-film and device characterization instruments, such as XRD (Powder diffraction), XRR (for thin-film characterization), Grazing-Incidence XRD (for the surface structure), AFM/MFM, Laser MOKE, and Vibrating Sample Magnetometer (VSM) for high- and low-temperature measurements.
3. Prepared operational manuals for electrical and physical property testing systems, including Quantum Design PPMS, DC/High-Frequency measurement setups, and fast-measurement LabVIEW VIs.
4. Conducted hands-on training for 10+ PhD students and undergraduates, covering instrument setup, calibration, measurement procedures, maintenance, and troubleshooting.
5. Designed instruction materials to ensure reproducible experimental results and safe cleanroom practices, enabling new users to independently operate complex fabrication and characterization systems.